

Problem Set 7 (two pages)

Statistics 24510-30040 (W25)

Due 9 am Thursday, March 6 (Note the unusual due day.)

Reminder: Final exam 10 am - 12 pm on Tuesday, March 11th.

Requirements

- Upload your pdf named "Lastname-Firstname-hw7.pdf" to Gradescope under Pset7. Be sure to mark the pages corresponding to each problem.
- Show your work. Provide detailed derivations and arguments in order to receive credit.
- You may discuss methods with others, but you should devise and write your answers by yourself completely and independently.

Problem assignments (Relevant reference: Rice Ch. 14, Casella & Berger Ch. 11.)

1. (Model mis-specification effects)

Suppose $Y_1, \dots, Y_n$ are *i.i.d.*, $Y_i \sim N(\beta_0, \sigma^2)$.

- (a) You fit the null model (the mean model) $Y_i = \beta_0 + \varepsilon_i$ and obtain $\hat{\beta}_0 = \arg \min_{\beta_0} \sum_{i=1}^n (Y_i - \beta_0)^2$.
- Find $\mathbb{E}(\hat{\beta}_0)$.
 - Find $\text{var}(\hat{\beta}_0)$.
 - Find $\mathbb{E}[(\hat{\beta}_0 - \beta_0)^2]$, the MSE (mean squared error).
- (b) You take the covariate variables $x_1, \dots, x_n$, fit a regression model $Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$, and obtain $\{\hat{\beta}_0, \hat{\beta}_1\} = \arg \min_{\beta_0, \beta_1} \sum_{i=1}^n (Y_i - \beta_0 - \beta_1 x_i)^2$.
- Find $\mathbb{E}(\hat{\beta}_0)$.
 - Find $\text{var}(\hat{\beta}_0)$.
 - Find $\mathbb{E}[(\hat{\beta}_0 - \beta_0)^2]$.

Attention: The data generating process is still *i.i.d.* Y_i 's with $Y_i \sim N(\beta_0, \sigma^2)$.

- (c) Comparison of results in (a) and (b).
- Are any or both of the estimators of β_0 unbiased?
 - Which estimate has large variance, or large MSE?
 - Discuss and comment on the comparison results.

2. (Random vectors and covariance matrices)

Consider a round table with n positions along it numbered from 1 to n in the clockwise direction from some arbitrary position. At each position a fair coin is tossed. To simplify the problem, here we consider the case $n = 5$.

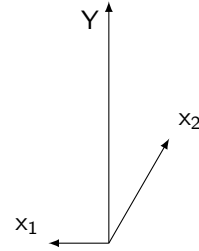
- (a) Consider the random vector $\mathbf{X}$ of length n whose i th component X_i is the indicator function of whether the toss at the i th position came up heads.
Find the mean vector $\mathbb{E}(\mathbf{X})$ and the covariance matrix $\text{Cov}(\mathbf{X})$.
- (b) Now consider the random vector $\mathbf{Y}$ obtained by, at each position, taking $\frac{1}{2}$ times the sum of the number of heads at that position (say, position i) and the next position (position $i + 1$).
Find the matrix A such that $\mathbf{Y} = A\mathbf{X}$.
- (c) Find the covariance matrix of $\mathbf{Y}$.
- (d) Now consider repeating this same averaging operation on $\mathbf{Y}$. That is, replace each component of $\mathbf{Y}$ by the average of itself and the component representing the next position at the table. Call this random vector $\mathbf{Z}$.
Find the covariance matrix for $\mathbf{Z}$.

3. (Projections of least squares method)

Vectors x_1, x_2 and Y are in $\mathbb{R}^d, d \geq 10$. Suppose that they fall in the same two-dimensional plane given by the surface of this page just as pictured, with lengths

$$\|Y\| = 1, \|x_2\| = 1/2, \|x_1\| = 1/4.$$

(Note that, the angle between Y and x_2 is about $\frac{\pi}{6}$.)



- If we regressed Y on just x_1 (no intercept), what would be the least squares estimate of the regression coefficient and the resulting residual sum of squares?
- Do the same as in (a) but using x_2 as the regressor.
- For the model in part (b), work out a good approximation to the t -statistic for the hypothesis that the regression coefficient is 0 and give your best guess for the associated p-value.
- When Y is regressed on both x_1 and x_2 (again no intercept), give reasonable approximations to the regression coefficients and the residual sum of squares. Explain how you obtained your answers.

4. (Prediction inference)

The linear model

$$Y \sim X\beta + \varepsilon, \quad \varepsilon \sim N_n(0, \sigma^2 I_n)$$

is fitted on n data observations $\{(y_i, x_{i1}, \dots, x_{i,p-1}), i = 1, \dots, n\}$. X is $n \times p$ with $\text{rank}(X) = p < n$.

For a new data point $x^* = [1 \ x_1^* \ \dots \ x_{p-1}^*]^T$, we can predict its mean response by $(x^*)^T \hat{\beta}$.

- Find the distribution of $(x^*)^T \hat{\beta}$.
- Construct a confidence interval for $\mu^* = (x^*)^T \beta$. Start by deriving a t test statistic.

5. (Central Limit Theorem and Variance Stabilization Transformation)

Consider the linear model $Y = X\beta + \varepsilon, \varepsilon \sim N(0_n, \sigma^2 I_n)$. An unbiased estimator for σ^2 is $\hat{\sigma}^2 = \frac{1}{n-p} \|Y - X\hat{\beta}\|^2$.

- Find the mean and variance of $\hat{\sigma}^2$ (no need of derivations).
- Use the CLT to find the asymptotic distribution of $\sqrt{n-p}(\hat{\sigma}^2 - \sigma^2)$ (i.e. when n is sufficiently large).
- Find a transform function g so that $\sqrt{n-p}(g(\hat{\sigma}^2) - g(\sigma^2)) \sim N(0, 1)$ asymptotically.
- Using the asymptotic result in part (c) to construct an approximate $(1 - \alpha)$ -confidence interval for σ^2 .